Financial Econometrics

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MZ Regression

To test the goodness of a prediction model: regress

$$Y_{t+h} = \alpha + \beta \widehat{Y}_{t+h} + u_t, \qquad \text{Test } H_0 : \ \alpha = 0, \ \beta = 1.$$

DM Test

To compare the accuracy of two prediction models. Let $L(\cdot)$ be the loss (ℓ^2, ℓ^1, etc.):

$$\ell_t^A = L\left(Y_{t+h} - \widehat{Y}_{t+h}^A\right), \qquad \ell_t^B = L\left(Y_{t+h} - \widehat{Y}_{t+h}^B\right).$$

Let $\delta_t = \ell_t^A - \ell_t^B$. Test $H_0 : \ \mathbb{E}\,\delta_t = 0$ with

$$DM = \frac{\bar{\delta}}{\sqrt{\widehat{\mathrm{Var}}(\bar{\delta})}}.$$

EMH

- **Weak form:** information from historical prices is fully reflected in the current price. Technical analysis is useless.
- **Semi-strong:** public information is fully reflected in the current price. Fundamental analysis is useless.
- **Strong form:** public & private information is fully reflected in the current price. Insider trading is useless.

Critiques of EMH

- *Grossman & Stiglitz (1980):* information collection and analysis is costly, so markets cannot be informationally efficient.
- *Shleifer & Vishny (1997):* arbitrage cannot always be possible or feasible, due to several kinds of limitations.
- *Joint-hypothesis critique:* to test the EMH we must set a model for the return sequence *ex ante*. When the null is rejected we cannot discriminate whether the return-sequence model or the EMH is the incorrect one.

RW Hypothesis

$$P_t = \mu + P_{t-1} + \varepsilon_t.$$

- **RW1:** ε_t i.i.d., $\mathbb{E}\,\varepsilon_t = 0$.
- **RW2:** ε_t independent over time, $\mathbb{E}\,\varepsilon_t = 0$.
- **RW2.5:** $\mathbb{E}(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0$.
- **RW3:** $\mathrm{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$.

Testing Methods

White-noise test (RW1).

$$Q_{BP} = T\sum_{j=1}^{p} \widehat{\rho}_j^2 \;\to\; \chi_p^2, \qquad Q_{LB} = T(T+2)\sum_{j=1}^{p} \frac{\widehat{\rho}_j^2}{T-j} \;\to\; \chi_p^2.$$

Time-series regression (RW1, RW2 with adjusted standard error).

$$X_t = \beta_0 + \beta_1 X_{t-1} + \cdots + \beta_k X_{t-k} + \varepsilon_t, \qquad \text{Test } H_0:\; \beta_1 = \cdots = \beta_k = 0.$$

Variance Ratio Under RW3 and homoskedasticity:

$$VR(q) = \frac{\mathrm{Var}(r_t + \cdots + r_{t-q+1})}{q\;\mathrm{Var}(r_t)} = 1.$$

If not,

$$VR(q) = 1 + 2\sum_{j=1}^{q-1}\left(1 - \frac{j}{q}\right)\rho(j).$$

Cowles–Jones Test (RW1)

$$\text{Continuations} = \#\{\, r_t \cdot r_{t-1} > 0 \,\}, \qquad \text{Reversals} = \#\{\, r_t \cdot r_{t-1} < 0 \,\};$$

they should be roughly the same.

Runs Test

$$\underbrace{+++}\;\underbrace{--}\;\underbrace{+}\;\underbrace{-}\;\underbrace{++} \qquad \Rightarrow \qquad 5 \text{ runs.}$$

#{runs} follows a certain distribution under RW2.

GARCH(p,q)

$$r_t = \underbrace{\mathbb{E}(r_t \mid \mathcal{F}_{t-1})}_{\mu_t} + \varepsilon_t, \qquad \varepsilon_t = \sigma_t e_t,$$

(μ_t usually modelled as an ARMA sequence.)

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{p} \alpha_i\,\varepsilon_{t-i}^2 + \sum_{j=1}^{q} \beta_j\,\sigma_{t-j}^2,$$

where (1) ε_t is strictly stationary; (2) e_t is an i.i.d. sequence, also independent of other r.v. (usually $e_t \sim N(0,1)$); (3) $\alpha_0 > 0$ and $\sum_i \alpha_i + \sum_j \beta_j < 1$.

Properties

(1) Define the martingale $v_t = \varepsilon_t^2 - \sigma_t^2$ with $\mathbb{E}_{t-1} v_t = 0$; then ε_t^2 is an ARMA sequence

$$\varepsilon_t^2 = \alpha_0 + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i)\,\varepsilon_{t-i}^2 + v_t - \sum_{j=1}^{q} \beta_j\,v_{t-j}.$$

(2) Conditional moments:

$$\mathbb{E}_{t-1}\varepsilon_t = \sigma_t\,\mathbb{E}_{t-1}e_t = \sigma_t\,\mathbb{E}e_t = 0, \qquad \mathrm{Var}_{t-1}\,r_t = \mathbb{E}_{t-1}\varepsilon_t^2 = \sigma_t^2\,\mathbb{E}_{t-1}e_t^2 = \sigma_t^2.$$

(3) Unconditional moments:

$$\mathbb{E}\,\varepsilon_t = \mathbb{E}\,\mathbb{E}_{t-1}\varepsilon_t = 0,$$

$$\mathbb{E}\,\sigma_t^2 = \alpha_0 + \sum_i \alpha_i\,\mathbb{E}\,\varepsilon_{t-i}^2 + \sum_j \beta_j\,\mathbb{E}\,\sigma_{t-j}^2 \;\Rightarrow\; \mathbb{E}\,\sigma_t^2 = \frac{\alpha_0}{1-\sum_i \alpha_i - \sum_j \beta_j}.$$

GJR-GARCH(p,o,q)

Conditional volatility is instead modelled as

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{p} \alpha_i\,\varepsilon_{t-i}^2 + \sum_{j=1}^{o} v_j\,\varepsilon_{t-j}^2\,\mathbb{I}(\varepsilon_{t-j}<0) + \sum_{l=1}^{q} \beta_l\,\sigma_{t-l}^2,$$

to model the *leverage effect*: larger variance after negative shocks.

EGARCH(p,o,q)

(where E stands for exponential)

$$\ln\sigma_t^2 = \alpha_0 + \sum_{j=1}^{p} \alpha_j \left(\left| \frac{\varepsilon_{t-j}}{\sigma_{t-j}} \right| - \sqrt{\tfrac{2}{\pi}} \right) + \sum_{i=1}^{o} v_i\,\frac{\varepsilon_{t-i}}{\sigma_{t-i}} + \sum_{l=1}^{q} \beta_l\,\ln\sigma_{t-l}^2.$$

New Impact Curve

$$m(e_t) = \sigma_{t+1}^2\big(e_t,\ \sigma_t^2 = \sigma_{t-1}^2 = \cdots = \tilde{\sigma}^2\big), \qquad \mathrm{NIC}(e_t) = m(e_t) - m(0).$$

Forecast

- **Method 1:** forecast with $\mathbb{E}_t\,\sigma_{t+h}^2$ recursively.
- **Method 2 (simulation):** construct B i.i.d. sequences of $\{e_t\}$; obtain B trajectories of $\widehat{\sigma}_{t+h}^2$, one per sequence.

Realized Variance → for High-Frequency Data

Basic estimand

$$RV_t^{(m)} = \sum_{i=1}^{m} \big(p_{it} - p_{i-1,t}\big)^2 = \sum_{i=1}^{m} r_{it}^2 \xrightarrow{m\to\infty} \int_t^{t+1} \sigma_s^2\,ds.$$

(r_{it} are within-day, time-stamped returns.) Assume $r_{it} \overset{iid}{\sim} N(\frac{\mu}{m}, \frac{\sigma^2}{m})$; then $\mathbb{E}\,RV_t^{(m)} = \sigma^2 + \frac{\mu^2}{m} \to \sigma^2$.

Potential bias — market microstructure noise

$$p_{it} = p_{it}^* + v_{it} \;\Rightarrow\; r_{it} = r_{it}^* + \eta_{it} \quad \text{(an MA(1))},$$

so $RV_t^{(m)} \approx RV_t^{*} + m\,\tau^2$, which *increases* with higher frequency. Solution (first-order autocovariance correction):

$$RV_t^{AC(1)} = \sum_{i=1}^{m} r_{it}^2 + 2\sum_{i=2}^{m} r_{it}\, r_{i-1,t}.$$

Modeling with Realized Variance

1. HAR model

$$RV_t \sim 1 + \underbrace{RV_{t-1}}_{\text{daily}} + \underbrace{\overline{RV}_{t-1}^{(5)}}_{\text{weekly}} + \underbrace{\overline{RV}_{t-1}^{(22)}}_{\text{monthly}}.$$

2. Realized GARCH

$$\begin{aligned} \text{return: } & \varepsilon_t = \sigma_t e_t, \quad e_t \sim N(0,1); \\ \text{GARCH: } & \log \sigma_t = \omega + \beta \log \sigma_{t-1} + v \log x_{t-1}; \\ \text{observation: } & \log x_t = \xi + \phi \log \sigma_t + \tau(e_t) + u_t, \quad u_t \sim N(0, \sigma_u^2). \end{aligned}$$

VaR — Value at Risk

The α-VaR is the largest loss on a portfolio with probability no more than α:

$$VaR = \sup_{VaR}\ \mathbb{P}(r_t \le -VaR) \le \alpha.$$

For the general conditional-variance model $r_{t+1} = \mu_{t+1} + \sigma_{t+1} e_{t+1}$ with $\mu_{t+1} = \mathbb{E}(r_{t+1} \mid \mathcal{F}_t)$, the conditional VaR at quantile α is

$$\mathbb{P}_t(\mu_{t+1} + \sigma_{t+1} e_{t+1} \le -VaR) = \alpha \ \Rightarrow\ VaR_t(\alpha) = -\mu_{t+1} - \sigma_{t+1}\, F^{-1}(\alpha).$$

CAPM

1. Global minimum variance

$$\min_w\ w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mathbf{1} = 1 \ \Rightarrow\ w^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}, \qquad \sigma^2 = \frac{1}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}.$$

2. Mean–variance frontier

$$\min_w\ w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mathbf{1} = 1,\ w^\top \mu = r.$$

The solution is $w^* = \lambda\, \Sigma^{-1}\mathbf{1} + v\, \Sigma^{-1}\mu$, where

$$A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}, \quad B = \mathbf{1}^\top \Sigma^{-1} \mu, \quad C = \mu^\top \Sigma^{-1} \mu, \quad \Delta = AC - B^2,$$

$$\sigma^2 = \frac{Ar^2 - 2Br + C}{\Delta},$$

the right branch of a hyperbola on the $\mathbb{E}(r)$–σ coordinate.

3. CML — Capital Market Line

$$\mathbb{E}\, r_p = r_f + \frac{\mathbb{E}\, r_m - r_f}{\sigma_m}\,\sigma_p.$$

Two-fund separation theorem

Any rational investor must hold a portfolio combined of 2 parts — the risk-free asset and the market portfolio — whose proportions are determined by the investor's preference for risk.

4. CAPM — Capital Asset Pricing Model

For any asset or portfolio p, its expected return and variance must satisfy

$$\mathbb{E}\, r_p - r_f = \beta_p\big(\mathbb{E}\, r_m - r_f\big), \qquad \beta_p = \frac{\mathrm{Cov}(r_p, r_m)}{\mathrm{Var}(r_m)}.$$

Proof. Construct a new portfolio p': λ in asset p and $(1-\lambda)$ in the market portfolio. By the two-fund separation theorem, p' attains the best Sharpe ratio at $\lambda = 0$, so

$$\left.\frac{dr_{p'}}{d\sigma_{p'}}\right|_{\lambda=0} = \frac{r_m - r_f}{\sigma_m}, \qquad \mathrm{LHS} = \frac{dr_{p'}/d\lambda}{d\sigma_{p'}/d\lambda},$$

with

$$\sigma_{p'} = \big(\lambda^2\sigma_p^2 + 2\lambda(1-\lambda)\sigma_{p,m} + (1-\lambda)^2\sigma_m^2\big)^{1/2}, \qquad r_{p'} = \lambda r_p + (1-\lambda) r_m.$$

Then

$$\left.\frac{d\sigma_{p'}}{d\lambda}\right|_{\lambda=0} = \frac{1}{\sigma_m}\big(\sigma_{p,m} - \sigma_m^2\big), \qquad \frac{dr_{p'}}{d\lambda} = r_p - r_m,$$

$$\Rightarrow \quad \frac{\sigma_m\,(r_p - r_m)}{\sigma_{p,m} - \sigma_m^2} = \frac{r_m - r_f}{\sigma_m},$$

which reduces to the CAPM. □

Notice: on the $\mathbb{E}\, r$–β coordinate, the CAPM equation plots the **Security Market Line (SML)**.

Factor Model

1. Fama–French

$$\mathbb{E}\, R_i - R_f = \beta_{i,\mathrm{MKT}}\big(\mathbb{E}\, R_m - R_f\big) + \beta_{i,\mathrm{SMB}}\,\mathbb{E}\, R_{\mathrm{SMB}} + \beta_{i,\mathrm{HML}}\,\mathbb{E}\, R_{\mathrm{HML}}.$$

Book-to-Market value (30%, 70% quantile)

Market value	High	Middle	Low
Small (50% q.)	S/H	S/M	S/L
Big	B/H	B/M	B/L

$$\mathrm{SMB} = \tfrac{1}{3}\big(\mathrm{S/H} + \mathrm{S/M} + \mathrm{S/L}\big) - \tfrac{1}{3}\big(\mathrm{B/H} + \mathrm{B/M} + \mathrm{B/L}\big),$$
$$\mathrm{HML} = \tfrac{1}{2}\big(\mathrm{S/H} + \mathrm{B/H}\big) - \tfrac{1}{2}\big(\mathrm{S/L} + \mathrm{B/L}\big).$$

2. Other factors

Momentum; Profitability (Robust Minus Weak / profitable-minus-unprofitable); ROE; rate of change on equity; ...

3. Fama–MacBeth two-step regression

- **Step 1 (time-series):** estimate risk exposure $\to \widehat{\beta}_i$ (i indexes the stock).

- **Step 2 (cross-sectional):** estimate risk price $\to \widehat{\lambda}_t$ (t indexes the time stamp).

GARCH-in-Mean (GARCH-M)

The conditional mean is allowed to depend on conditional volatility, so that a *risk premium* enters returns directly:

$$r_t = \mu + \lambda\,\sigma_t^2 + \varepsilon_t, \qquad \varepsilon_t = \sigma_t e_t, \quad e_t \sim N(0,1),$$
$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{p} \alpha_i\,\varepsilon_{t-i}^2 + \sum_{j=1}^{q} \beta_j\,\sigma_{t-j}^2.$$

(One may use $\lambda\sigma_t$ or $\lambda \log \sigma_t^2$ in place of $\lambda\sigma_t^2$.) Idea: λ is the price of risk. If $\lambda > 0$, periods of higher conditional variance command a higher expected return — a time-varying risk premium. $\lambda = 0$ collapses to the plain GARCH mean. Estimation: by MLE jointly with the variance parameters.

Realized GARCH$(1,1)$ — Model & Estimation

A GARCH whose variance recursion is driven by a *realized measure* x_t (e.g. RV_t), linked to σ_t^2 through a measurement equation.

Model

$$\begin{aligned}
\text{return:}\quad & r_t = \sigma_t e_t, \qquad e_t \sim N(0,1);\\
\text{GARCH:}\quad & \log\sigma_t^2 = \omega + \beta\log\sigma_{t-1}^2 + \gamma\log x_{t-1};\\
\text{measurement:}\quad & \log x_t = \xi + \phi\log\sigma_t^2 + \tau(e_t) + u_t, \qquad u_t \sim N(0,\sigma_u^2),
\end{aligned}$$

with leverage function $\tau(e) = \tau_1 e + \tau_2(e^2-1)$ (so $\mathbb{E}\,\tau(e) = 0$), capturing the asymmetric response of volatility to the sign of returns.

Estimation (MLE)

The measurement equation lets the realized measure enter the likelihood, so the joint log-likelihood factors into a return part and a measurement part:

$$\log L = -\frac{1}{2}\sum_t\left(\log\sigma_t^2 + \frac{r_t^2}{\sigma_t^2}\right) - \frac{1}{2}\sum_t\left(\log\sigma_u^2 + \frac{u_t^2}{\sigma_u^2}\right) + \text{const.}$$

Maximize over $(\omega,\beta,\gamma,\xi,\phi,\tau_1,\tau_2,\sigma_u^2)$ by recursively filtering σ_t^2 from the GARCH equation. Adding the realized measure typically sharpens volatility estimates relative to a return-only GARCH.

DCC Model (Dynamic Conditional Correlation, Engle 2002)

A parsimonious way to model a *time-varying* covariance matrix H_t of an n-dimensional return vector, separating variances from correlations:

$$H_t = D_t\,R_t\,D_t, \qquad D_t = \operatorname{diag}\big(\sqrt{h_{1t}},\ldots,\sqrt{h_{nt}}\big),$$

where the h_{it} come from univariate GARCH models and R_t is the conditional correlation matrix.

Two-step estimation

- **Step 1:** fit a univariate GARCH to each series; obtain h_{it} and the standardized residuals $z_{it} = \varepsilon_{it}/\sqrt{h_{it}}$.

- **Step 2:** let the standardized residuals drive a dynamic "quasi-correlation" recursion

$$Q_t = (1 - a - b)\,\bar{Q} + a\,z_{t-1}z_{t-1}^\top + b\,Q_{t-1},$$

  with $\bar{Q} = \mathbb{E}\big(z_t z_t^\top\big)$ the unconditional covariance of the standardized residuals, and normalize to a valid correlation matrix

$$R_t = \widetilde{Q}_t^{-1/2}\,Q_t\,\widetilde{Q}_t^{-1/2}, \qquad \widetilde{Q}_t = \mathrm{diag}(Q_t).$$

Parameters $a, b \geq 0$ with $a + b < 1$ ensure stationarity and a mean-reverting correlation. Step 1 and Step 2 are estimated sequentially by (quasi-)MLE.

Filtered Historical Simulation (FHS)

A semi-parametric VaR method: use a volatility model to *filter* returns, then bootstrap the empirical innovations (no distributional assumption on tails).

- **Filter:** fit a volatility model (e.g. GARCH) to returns; obtain $\widehat{\sigma}_t$ and standardized residuals $\widehat{z}_t = \varepsilon_t/\widehat{\sigma}_t$.
- **Empirical innovations:** treat $\{\widehat{z}_t\}$ as the innovation distribution — this preserves fat tails and asymmetry.
- **Forecast & resample:** compute $\widehat{\sigma}_{t+1}$ from the recursion; draw z^* with replacement from $\{\widehat{z}_t\}$ and form simulated returns

$$r_{t+1}^* = \widehat{\mu}_{t+1} + \widehat{\sigma}_{t+1}\,z^*.$$

  For an h-day horizon, iterate the variance recursion forward using the sampled innovations to build return paths.
- **VaR:** the empirical α-quantile of the simulated returns,

$$VaR_{t+1}(\alpha) = -\,\widehat{\mu}_{t+1} - \widehat{\sigma}_{t+1}\,\widehat{q}_\alpha,$$

  with $\widehat{q}_\alpha$ the α-quantile of $\{\widehat{z}_t\}$.

Idea: GARCH captures volatility clustering; the empirical innovations capture non-normal tails — combining the strengths of parametric and historical approaches.

Evaluating VaR (Backtesting)

Define the violation ("hit") indicator

$$I_t = \mathbb{I}\big(r_t < -VaR_t(\alpha)\big).$$

Under a correct model, $I_t \overset{iid}{\sim} \mathrm{Bernoulli}(\alpha)$.

Absolute assessment — does a single model pass?

- **Unconditional coverage (Kupiec, POF).** With $x = \sum_t I_t$ violations in T days and $\widehat{\pi} = x/T$, test $H_0:\ \widehat{\pi} = \alpha$ via

$$LR_{uc} = -2\log\frac{\alpha^x(1-\alpha)^{T-x}}{\widehat{\pi}^x(1-\widehat{\pi})^{T-x}} \;\to\; \chi_1^2.$$

- **Independence (Christoffersen).** Violations should not cluster; test $\{I_t\}$ for serial independence, $LR_{ind} \to \chi_1^2$.
- **Conditional coverage.** Joint test of correct rate *and* independence, $LR_{cc} = LR_{uc} + LR_{ind} \to \chi_2^2$.

Relative assessment — which of two models is better?

Score each model with a loss function and compare; e.g. the tick / pinball (quantile) loss

$$L_t(\alpha) = \big(\alpha - I_t\big)\big(r_t + VaR_t(\alpha)\big),$$

where lower average loss is better. Compare the two models' average losses with a Diebold–Mariano test on the loss differential — linking back to the DM test at the top.